Mechanistic Interpretability of Open Source LLMs

Reproduction of On the Biology of a Large Language Model

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Motivation and Scientific Justification

A large language model computes by transforming a vector with several thousand coordinates through a long sequence of layers, and everything the model has extracted from its input at a given moment is encoded in this vector, the hidden state. Understanding how human interpretable structure is organised in this space is a central problem of the field. It determines whether the internal state of a model can be audited for knowledge and for safety relevant properties, whether behaviour can be controlled by direct intervention on the state, and more broadly whether the model's computation can be studied as an object of science rather than treated as a black box. The working hypothesis underlying much of this programme is geometric. Concepts are taken to correspond to directions in the hidden state space, so that the presence of a concept can be read as a projection onto the right axis. The most widely used operational tool built on this hypothesis is the linear probe, a simple classifier trained on recorded hidden states, and when a probe detects a concept with high accuracy the concept is declared to be represented along the direction the probe found [3]. Applications then identify this direction with the mechanism. Audits read it, and steering methods intervene along it. The premise underneath is that the direction from which a concept can be read is also the direction through which the model uses the concept when producing its answer. This project examined the internal geometry of one concept in full, from the axes that carry it to the features that write it, and found that the premise fails in a precise and explainable way.

Methodology

A controlled task with an unambiguous ground truth was chosen so that cause and effect could be measured at a single position. The model reads a short description of a nuclear decay process, for example which particle is emitted and how the charge of the nucleus changes, and answers whether the decay is alpha or beta. The answer is one token and physics fixes the truth. The model is an open four billion parameter network with thirty six layers, equipped at every layer with a pretrained dictionary that decomposes the layer's computation into thousands of sparse features, many of which activate on recognisable properties of the input.

The project began by reproducing the circuit tracing methodology of Anthropic [1, 2] on this open model. Two quantitative observations from the reproduction motivated the move to the level of directions. First, removing any single feature or pathway left the behaviour essentially unchanged, while removing feature ensembles reproduced the behavioural effect, so the causal structure was carried collectively rather than by nameable components. Second, the features with the largest measured contribution to the answer were largely disjoint from the features whose activations best detected the concept. Both facts are properties of the hidden state as a whole rather than of its individual components, so the analysis turned to directions of the hidden state, measured independently at every layer (Figure 1). The reading direction is the axis of the optimal linear classifier of the concept, and it is measured because it is exactly what a probe returns, so it represents what probing can see. The writing direction is the difference the model deposits into the hidden state between the two classes, the trace the concept leaves. The use direction is the axis along which a small displacement of the hidden state most changes the answer, the sensitivity of the output. If a concept were a single direction, the three measurements would find the same axis.

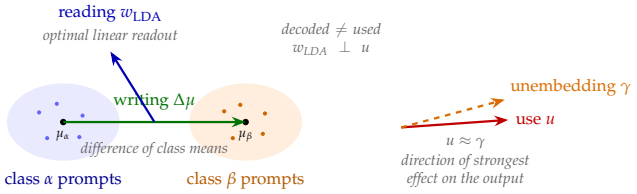


Figure 1: The four geometric directions. The reading direction is the optimal linear readout of the class, the writing direction is the difference of the class means, the use direction is the direction of the strongest effect on the output, and toward the output it aligns with the unembedding contrast through which the answer is produced. The central finding is that the reading and use directions are nearly orthogonal.

Key Findings

The three directions are distinct and nearly mutually orthogonal, and the difference is causal. The reading direction detects the concept almost perfectly, yet displacing the hidden state along it leaves the answer unchanged even at large intensities, while the same displacement along the use direction reliably flips the answer. Detectability and causal relevance are different properties. The dissociation is not an accident of one dataset. On a second family of concepts, where the reading direction lies closer to the use direction, the same intervention does flip the answer, so causal relevance tracks the geometric alignment with the use direction.

The orthogonality is explained rather than observed. An exact identity shows that an optimally fitted probe is attenuated along any axis that carries a large variance within each class, in direct analogy with a matched filter that turns away from channels with a loud background. The use direction is such an axis because the fluctuations of the model's confidence from example to example lie along it. The very fact that the model uses a direction fills that direction with variance, and an optimal reader is therefore expected to turn away from it. The failure of the premise is a consequence of the statistics of the task, derived in closed form and

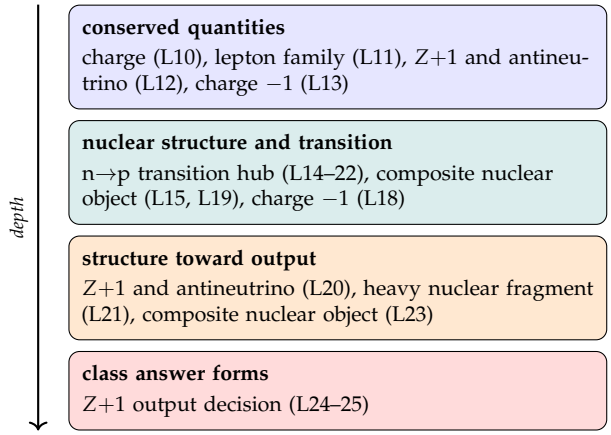


Figure 2: The alpha and beta decay circuit organised by depth. Clusters of features enriched for physical properties of the decay enter in sequence, conserved quantities in the early layers, nuclear structure and the transition in the middle layers, and the class answer only at the end. The class is computed from a physical description rather than stored as a label.

confirmed numerically layer by layer.

The content of what the model writes was identified through the feature dictionaries. The features that carry the written difference detect physical properties of the decay process rather than the class, the emitted charge, the lepton number, the composition of the nucleus, the type of transition, and they activate almost equally on both classes. Grouped by shared causal role, these features form clusters that enter the computation in a physically ordered sequence across depth, conserved quantities first, nuclear structure and the transition next, and the class decision only in the late layers (Figure 2). The model does not store the answer as a label. It builds a physical description of the process and assembles the class from that description toward the output.

Future Steps

Four extensions follow directly. Testing the three way separation on other model families establishes generality. Replacing the linear reader with nonlinear probes checks that the gap is not an artifact of linearity. Measuring the realised contribution of each feature to the writing direction strengthens the expressibility result into an account of the write itself. Manipulating the task so that surface shortcuts are opened or closed tests when the stored content is actually recruited by the computation and turns the present geometric picture into a dynamical one.

Research Impact

For the practice of model auditing the result gives a concrete calibration. Detecting a concept certifies what a model stores, not what it uses, so audits built on probes and steering methods built on probe directions can systematically miss the machinery of the behaviour they target. The constructive part of the result is that the gap is not random. It is geometric, predictable from measurable quantities, and it comes with a derived mechanism, which turns the distinction between reading, writing, and use into a practical checklist for any analysis that treats a concept as a single direction. The project also contributes a controlled and fully reproducible case study, with every reported number traceable to a committed result file, showing that the internal organisation of a language model on a physics task follows the physics of the process rather than the labels the task asks it to produce.

References

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